Sai Guntupalli

Senior Data Engineer
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PROFESSIONAL SUMMARY

Senior Data Engineer with around 10 years of experience building scalable backend systems and data-intensive applications. Current work at Disney focuses on combining core Data Engineering with Agentic AI to automate manual workflows. Specializes in the Python ecosystem, PySpark, and cloud architectures (AWS), with a strong background in the Media and Finance sectors.

CERTIFICATIONS

- NVIDIA Generative AI LLMs Certification
- GCP Professional Data Engineer
- Databricks Certified Data Engineer Associate

TECHNICAL SKILLS

- **Languages:** Python, SQL, Scala, Java, JavaScript, Shell Scripting.
- **AI & Automation:** Agentic AI (LangChain, LangGraph), GitHub Copilot, AWS Bedrock, Amazon Q, Databricks Genie, HuggingFace, OpenAI API.
- **Data Science & Analysis:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn.
- **Data Engineering & Cloud:** PySpark, Snowflake, Databricks, Apache Airflow, Kafka, Flink, AWS (S3, Lambda, EMR, Kinesis, Glue, Step Functions).
- **DevOps & Tools:** Docker, Jenkins, GitHub Actions, GitLab CI/CD, Terraform.
- **Databases:** Snowflake, PostgreSQL, MySQL, MongoDB, SQLite.

PROFESSIONAL EXPERIENCE

THE WALT DISNEY COMPANY | **Senior Data Engineer** | *Aug 2022 – Present*

Marketing Mix Modeling (MMM)

- Owned the end-to-end data platform powering econometric models that optimize $100M+ in annual marketing spend across Disney+, Hulu, ESPN+, and ESPN.
- Automated ingestion pipelines using AWS Lambda functions to feed paid media spend and performance data into the MMM warehouse.

- Built configurable data quality (DQ) checks to validate accuracy and completeness across all ingested sources.
- Built a monitoring system to track ingestion job status and surface pipeline health in real time.

Event Level Attribution Platform (ELAP)

- Built a web attribution system tracking user journeys from ad click through subscription conversion for Disney+ and ESPN+.
- Engineered and optimized ETL pipelines with complex business logic to reliably process millions of event records daily.

Project Armory

- Led the re-architecture of ESPN's marketing data pipeline, consolidating 15 legacy notebooks and 6 source systems into a single unified pipeline.
- Designed the data model and standardized schema behind a unified view of ESPN marketing performance.
- Partnered extensively with business stakeholders to align the new pipeline with reporting and modeling needs.

ETL Framework & Platform Standardization

- Built a comprehensive framework standardizing ETL creation and data source management across the team's pipelines.
- Designed the framework to be config-driven, enabling integration with a wide range of data sources — including Snowflake, Databricks, S3, and Google Sheets — without requiring custom code for each new source.
- Included a config-driven data quality (DQ) framework, enabling consistent validation across all pipelines.

AI Developer Tooling & Skills Framework

- Took the initiative to build a standardized framework for creating and using AI-assisted development skills across the team.
- Built a Databricks optimizer skill that analyzes cluster configs and recent run logs to identify bottlenecks and recommend optimized cluster configurations.
- Built PR creation, Jira story creation, and Project Armory migration and validation skills — one converts legacy notebooks into the new unified pipeline format, another validates new code against legacy logic and business requirements.
- Cut manual effort significantly across migrations, code reviews, and story creation by standardizing these workflows.

AirBoard

- Built a DAG monitoring dashboard organizing the team's Airflow pipelines by project, with execution status, run history, SLA compliance, and lineage dependencies between DAGs.
- Consolidated OnCall runbooks and DAG-level information, giving OnCall engineers a single place to manage all pipelines.
- Built a Business Leadership view surfacing high-level pipeline health metrics for non-technical stakeholders.
- Integrated with PagerDuty to link DAGs directly to incidents.

NBCUNIVERSAL | **Senior Data Engineer |** *Oct 2021 – July 2022*

Total Audience Measurement (TAMI)

- Worked on a measurement tool that tracks viewing consumption across various demographics and genres.
- Optimized Spark pipelines using Python, improving data processing speeds by 70%.
- Automated manual data workflows by building serverless functions with AWS Lambda and orchestrating them via AWS Step Functions.

Revenue Dashboard (RevDash)

- Re-engineered legacy data pipelines, reducing total execution time from 12 hours to 3 hours.
- Migrated the existing environment to Spark 3, utilizing its updated optimization features to handle larger data volumes with smaller cluster footprints.
- Converted complex business logic from Python to Spark Scala to meet high-throughput performance requirements.

EXPERIAN | **Data Engineer |** *Jan 2019 – Sept 2021*

- Built data pipelines for USAA Bank to process credit bureau data and identify customer eligibility for various financial products.
- Used Apache Airflow to schedule and manage Spark jobs, ensuring consistent data delivery for analytical modeling.
- Wrote Python and Shell scripts to automate cluster monitoring and data validation tasks, reducing manual oversight by 25%.
- Collaborated with data scientists to optimize feature engineering pipelines, making them more modular and easier to maintain.

TWITCH | **Data Engineer** | *July 2018 – Dec 2018*

- Developed a real-time alerting system that monitored streaming data and engagement metrics.

- Used Apache Flink and AWS Kinesis to process high-velocity data streams with low latency.
- Built Python-based AWS Lambda functions to send immediate notifications to Slack and email when specific data thresholds were breached.

EXPERIAN | **Software Engineer** | *May 2017 – June 2018*

- Worked on the CrossCore integration platform, focusing on connecting third-party fraud detection services.
- Used Java to develop backend services and managed data mapping between systems using XML and JSON.
- Focused on writing clean, tested code and participated in regular peer code reviews to maintain system stability.

TAVANT | **Project Trainee** | *Oct 2016 – Apr 2017*

- Completed a rigorous training program focused on Python, Java, and the Hadoop ecosystem (Hive, Sqoop, Spark).
- Worked on simulated real-world projects to gain practical experience in data ingestion and transformation.

EDUCATION

Bachelor of Technology in Computer Science and Engineering (2012 - 2016)